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ORIGINAL RESEARCH

Secure Model for Dynamic Access Control and Unreliable Access Point Detection: Enhancing QoS Through SDN in Wireless Networks

Hafsa Ait Oulahyane[1] · **Ayoub Bahnasse**[2] · **Assia Bakali**[3] · **Said Broumi**[1] · **Ibrahim M. El-Hasnony**[4] · **Mohamed Talea**[1]

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Abstract
Over the past few years, we have seen the spread of wireless networks in institutions, businesses, and private homes. Their wide variety, market availability, and ease of implementation have made this wireless technology more popular than wired networks. Nevertheless, enterprises are still demanding better quality of service to meet the needs of sensitive applications and avoid data loss. The SDN approach provides the ability to manage QoS dynamically via a controller. In this paper, we developed a secure model based on dynamic access control and the detection of unreliable access points. We also proposed QoS management in a wireless network by adopting the SDN approach to address connection problems and efficiently utilize resources. By employing these tools and techniques, we can narrow down all possible options to one single best choice. By comparing the results, our model validates its effectiveness in minimizing latency, jitter, packet loss, and transaction delay and offers a good opinion score.

Keywords Wireless network · Access control · Secured access point · Quality of service · SDN · Packet loss · Latency · Jitter · Delay · MADM

Introduction

The ever-evolving needs of today's users have surpassed the capabilities of traditional networks, prompting the emergence of a revolutionary network concept called Software Defined Networking (SDN). In 2006, a research group from Berkeley University and Stanford introduced SDN as an innovative approach to network architecture [1–3]. SDN disrupts traditional networks by providing network administrators with dynamic control and management of network resources through programmable interfaces.

The fundamental principle of SDN involves decoupling the control plane from the physical plane (data plane) in network infrastructure. This separation brings forth a range of advantages and benefits, including:

- *Increased reliability and scalability*: SDN enables improved reliability by centralizing control and management, allowing for efficient allocation and utilization of network resources. Additionally, it provides scalability to adapt to changing network demands and growth requirements.
- *Enhanced flexibility and agility*: With SDN, network administrators gain the ability to dynamically adjust network behavior and policies through software-based programmability. This flexibility allows for quick provisioning, deployment, and reconfiguration of network services, leading to enhanced agility and adaptability.
- *Centralized management and global network view*: SDN provides a centralized management platform that offers a holistic view of the entire network infrastructure. This centralized control allows administrators to have a comprehensive understanding of network operations, moni-

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Hafsa Ait Oulahyane
hafsa.lahyane@gmail.com

1 Laboratory of Information Processing, Faculty of Science Ben M'Sik, University Hassan II, Casablanca, Morocco

2 Ensam Casablanca, University Hassan II, Casablanca, Morocco

3 Naval Royal School, Casablanca, Morocco

4 Faculty of Computers and Information, Mansoura University, Mansoura, Egypt

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tor traffic, and implement policies consistently across the network.

- *Open and neutral standards*: SDN promotes open standards, making it vendor-agnostic and enabling interoperability among different network devices and solutions. This openness fosters innovation, encourages collaboration, and avoids vendor lock-in, ultimately benefiting end-users.

The SDN paradigm has revolutionized network management and brought about significant advancements in addressing the evolving requirements of modern networks. By offering increased reliability, scalability, flexibility, agility, centralized management, and adherence to open standards, SDN has become a pivotal technology in shaping the future of networking.

The SDN architecture is based on three layers as described in Fig. 1 below:

- *The application layer*: represents a set of programs called "SDN applications" that communicate their network resource requirements to the controller through the northbound interface.
- *The control layer*: includes NBI agents, SDN control logic, and Control of the Data-Plane Interface (CDPI) driver. It centralizes the intelligence of the network and gives a global view of the network via the southbound interface.
- *The infrastructure layer*: as its name suggests, the layer collects data and handles its transmission. It is also called the physical layer because it is the layer on which virtualization will be established.

Heterogeneous Networks

In response to the skyrocketing demand for mobile data traffic, heterogeneous networks have emerged as a viable solution [4, 5]. These networks cater to the evolving needs of users who no longer solely rely on voice services but also require advanced services like video streaming. Heterogeneous networks enable the coexistence of different network technologies, satisfying the increasing demand for throughput while allowing for the continuous evolution of technologies without the need to replace existing ones. Prominent technologies in this domain include WiFi, WiMAX, UMTS, LTE, and 5G [6, 7]. In this article, our focus will be on WiFi technology.

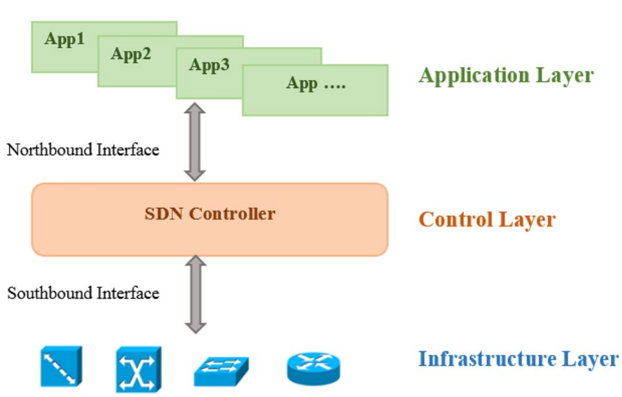


Fig. 1 Software-defined Network architecture

WiFi, as one of the widely adopted standards for wireless network deployment, facilitates communication between multiple devices using radio frequency. The IEEE 802.11 standard, available at speeds ranging from 1 to 2 Mbps [8], serves as the foundation. To enhance performance aspects such as range and throughput, this standard has undergone several iterations, resulting in various versions [8]. Two primary modes of operation are supported by WiFi: ad hoc mode for establishing peer-to-peer networks, and infrastructure mode for establishing wired networks using access points.

The versatility of WiFi technology has contributed to its widespread adoption and usage in various settings, including homes, offices, public spaces, and educational institutions. It provides users with convenient wireless connectivity, enabling seamless communication and data exchange between devices. The continuous advancements and improvements in WiFi standards have further bolstered its capabilities, ensuring higher data rates, improved coverage, and enhanced user experience.

Handover and SDN

The emergence of heterogeneous networks has enabled users to seamlessly transition between different networks, a process known as handover [9]. A smooth and efficient handover is crucial to maintain the quality of service and user experience. Software-defined networking (SDN) technology plays a vital role in managing mobility effectively.

In the context of 5G mobile networks, several works have focused on mobility management using the SDN paradigm. Meneses [10] proposed an SDN-based framework as a mobility management protocol, evaluating its performance based on handover delay and the impact of different threshold configurations in network and mobile-initiated scenarios. Bilen [11] addressed the handover delay issues caused by ultra-densification by proposing a mobility strategy that leverages SDN and estimates the available resources. The proposed solution selects optimal eNBs (evolved Node Bs) by considering their transition probabilities and available resource probabilities using a Markov chain formulation. Gharsallah [12] also tackled the handover problem by proposing an optimized approach for vertical handover selection, cooperating with media-independent handover (MIH) and SDN to achieve transparent vertical handovers while ensuring a high-quality of service.

Multiple Attribute Decision-Making (MADM) is a method used to determine the most preferable choice among a set of alternatives based on various criteria. SAW (Simple Additive Weighting), Weighting Product (WP), and TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) are three existing MADM approaches commonly employed to address decision-making challenges. In the context of handover, researchers have utilized different approaches. For instance, SAW and WP were used in [14] to rank visited networks and select the best one. In [15], TOPSIS was employed to ensure a smooth transition, with the goal of selecting a target network that is closest to the optimal solution while being the furthest from the worst possible network.

SDN and Security

The security of a network is a paramount concern for users today, and it encompasses several important properties. These properties include authentication, confidentiality, integrity, non-repudiation, and availability.

- Authentication ensures secure user access by requiring proof of identity, ensuring that only authorized individuals can access the network.
- Confidentiality protects user data from unauthorized access and ensures that it remains private.
- Integrity guarantees that data is not tampered with or lost during transmission over the network, ensuring that it retains its original state.
- Non-repudiation ensures that neither party can deny a transaction or action that has taken place on the network, providing accountability.
- Availability refers to the network's ability to perform its functions at any time and under specific conditions. It depends on the frequency of failures and the time required to recover from those failures.

The SDN architecture has made significant advancements in network security, particularly considering the heterogeneity of networks and the increased rate of handovers. While some initially expressed concerns that the use of a controller might facilitate attacks, studies have shown that the SDN architecture actually improves attack detection in a network. It allows for cost-effective monitoring of network traffic to identify anomalies that may indicate a network attack [16–18]. Moreover, it enables swift network modifications to mitigate attacks once identified. Additionally, encryption plays a significant role in protecting data both in transit and at rest.

The emergence of SDN in the last few years has shaken up the networking world with new solutions that address quality of service, scalability, reliability, and security. The adoption of SDN and its application on mobile and wireless networks was a remarkable transition but it had several challenges:

- Security
- The use of resources
- The guarantee of the quality of service

Our model combines two key features dynamic access control and untrusted access point detection. Dynamic access control refers to the ability to dynamically adjust the security level of a particular device or user based on their behavior and the data they access. This allows for a more flexible and adaptive security system. Detecting untrusted access points is also an important part of our security model. Untrusted access points can pose a significant threat to the security of a network, as they can potentially allow unauthorized access or spread malware. Our model uses advanced algorithms to detect and isolate these untrusted access points to ensure that they do not pose a threat to the network. In addition to security, we also realize the importance of Quality of Service (QoS) management in a wireless network. To address connection issues and ensure the efficient use of resources, we proposed the adoption of a software-defined networking (SDN) approach. This approach enables centralized control and management of the network, which facilitates monitoring and resource allocation based on need.

And here are some features of our model:

- Remedy the problems of connecting to a wireless network within an institution
- Provide maximum security to the network
- Evaluate the quality of service in a permanent way
- Efficiently use resources.

Using this structure, let us look at the rest of the document: Sect. "Related works" provides a brief overview of relevant literature, Sect. "MADM Methods" provides a review of MADM methods, Sect. "Secure Model for Dynamic Access Control and Unreliable Access Point Detection" outlines our proposed approach methods, "Results and Discussion" introduces experimental findings, and "Conclusion" concludes the paper by discussing potential future directions.

Related Works

SDN and Wifi

The increasing demand for data transmission due to the popularity of smartphones and tablets has put a strain on cellular networks, leading to capacity limitations. To alleviate this, WiFi access points have been deployed to enable high-speed localized data transmission. This technology has continued to evolve and has integrated the new architecture of software-defined networking (SDN). According to Honglin Hu's essay [19], Software-defined wireless networking (SDWN) offers a cost-effective solution to address emerging needs such as adaptability and vendor independence. SDWN separates the data and control planes, allowing network controllers to be programmed directly and abstracting the complexities of the wireless infrastructure.

While SDN and network virtualization have predominantly been applied to wired networks, Mao Yang [20] explores their application to wireless networks as well. Wireless networks pose unique challenges and opportunities, and Software-Defined Wireless Networking (SDWN) along with Wireless Network Virtualization (WNV) hold promise in addressing these issues. The article focuses on SDWN and WNV, highlighting their key concepts, benefits, ongoing research, essential technologies, and existing challenges. By leveraging SDWN and WNV, this article demonstrates how these technologies can contribute to the advancement of future mobile and wireless networks, offering potential solutions to the pressing issues faced by Mobile Wireless Networks (MWN).

SDN Wireless Qos

Ensuring quality of service (QoS) in mobile and wireless networks has become a crucial requirement for users. To address this demand, researchers have applied the innovative Software-Defined Networking (SDN) architecture to these networks, resulting in various proposed mechanisms and architectures.

In the paper by Rafid Mustafiz [21], the Spanning Tree Protocol (STP) is employed in an SDN-based wireless network. The study analyzes QoS using Mininet-Wifi, focusing on parameters such as bandwidth utilization, packet transmission rate, round trip time, maximum achieved throughput, packet loss rate, and delay time. Different base stations within the SDN architecture are evaluated to observe the performance of these QoS parameters.

Dibakar Das [22] introduces a novel mechanism for QoS negotiation between flows traversing a wired domain and a wireless domain, each controlled by separate SDN controllers. The paper defines a generic mechanism for mapping QoS parameters and develops utilities for each domain based on these parameters.

Azeddine KHIAT [23] proposes an extensible SDN architecture for intelligent, dynamic, and adaptive QoS management in both heterogeneous and homogeneous wireless networks. The simulation results demonstrate the effectiveness of this architecture in various applications, outperforming traditional solutions.

In the context of improving QoS, network operators have explored the utilization of different radio access technologies (RATs) in a heterogeneous manner to enhance service quality for subscribers. Nayak [24] presents the application of network virtualization and SDN architecture in managing heterogeneous technologies within wireless networks. The paper proposes an SDN/NFV-based architecture for unified control and management of heterogeneous RATs in 5G wireless networks.

SDN Security Wireless

The networking industry has undergone significant changes with the adoption of Software-Defined Networking (SDN), particularly in the context of mobile and wireless networks. As user demands continue to rise, there is a need to address the limitations of existing network architectures in terms of management and security. Ding [25] explores the security aspect in mobile and wireless networks, providing an in-depth analysis of existing problems and proposing potential solutions.

In the realm of wireless network security, Yanyan Han [26] presents a new framework for managing authentication and access control in wireless environments within the SDN paradigm. This approach addresses insecure access and latency issues while ensuring optimal access to wireless networks.

With the emergence of the Internet of Things (IoT), smart home systems have gained popularity, leading to new vulnerabilities at the network layers. Soteris Demetriou [27] introduces HanGuard, a control technique that leverages SDN architecture to enhance the security of communication between IoT devices and smart home systems. By utilizing a user-space monitoring application and control plane decisions, HanGuard effectively safeguards IoT devices from unauthorized access.

SDN Handover Wireless

In the context of mobile and wireless networks, the adoption of SDN technology has been instrumental in addressing challenges related to traffic overload and mobility management in dense and heterogeneous environments. Several research works have focused on applying SDN in these networks and proposing solutions to optimize the handover process.

Chen [28] presents a mobility management scheme called Mobility-SDN, which operates at the layer 2 level within an SDN enterprise network. This scheme prepares handovers in parallel through address resolution, effectively reducing the pause time caused by handovers.

Meneses [13] explores different handover scenarios in mobile and wireless networks, considering handovers initiated by either the mobile or the network. The study evaluates the handover delay and threshold as key parameters in an SDN-based framework used for mobility management.

Zeljkovic [29] introduces an SDN-based Wi-Fi framework that employs four algorithms for handover decision-making. These algorithms consider parameters such as received signal strength, location, access point load, and mobility model to select the optimal access point for handover.

Sun [30] focuses on achieving seamless handover and maintaining a better quality of service in wireless networks. The proposed SDN-based handover management scheme utilizes parameters like received signal strength, RSS prediction, and available bandwidth. A multi-attribute fuzzy logic-based handover algorithm is employed to evaluate network performance and make informed handover decisions.

Table 1 provides a summary of the related work in this area, highlighting the different approaches and parameters considered by each study.

MADM Methods

Techniques that use multiple attributes (MADM) can be used to rank and pick the optimal network based on certain quality of service factors.

The coexistence of heterogeneous networks with varied features and architectures shows the wide range of wireless access methods. Depending on the quality of service (QoS) characteristics and the traffic classifications of mobile users, this variety provides mobile terminals with a range of connectivity possibilities. The necessity for seamless mobile terminal connectivity transfer from one network to another arises as a result of this increased adaptability. The term "vertical handover" refers to when this procedure is carried out between two networks that have different architectures and air interface protocols (VHO). Phases of the vertical handover process include the handover initiation phase, the handover decision phase, and the execution phase [43].

To select the best network for vertical handover execution, numerous performance criteria are taken into account in the decision-making process. To analyze a group of competing choices, multiple-attribute decision-making (MADM)

Table 1 A comparison summary of the QoS management approach

Criteria	Contributions			
Throughput	Reference	Van Adrichem, Niels Lm, Doerr, Christian, Kuipers, Fernando A	He Qiang, Shengbao Wang	He Qiang, Xingwei Wang, and Min Huang
	Approach	OpenNetMon [31]	OpenLL [35]	OpenLh SDN [39]
	Year	2014	2017	2018
Latency	Reference	M. Selmchenko, M. Beshley, O. Panchenko, And M. Klymash	S. Rowshanrad, S. Namvarasl, And M. Keshtgari	X. Zhang, W. Hou, L. Guo, S. Wang, Q. Zhang, P. Guo, And R. Li
	Approach	monitoring system for end-to-end packet delay measurement in software-defined networks [32]	Queue Monitoring System in OpenFlow Software Defined Networks [36]	Joint optimization of latency monitoring and traffic scheduling [40]
	Year	2016	2017	2018
Bandwidth	Reference	H. Tahaei, R. Salleh, S. Khan, R. Izard, K. K. R. Choo, And N. B. Anuar	Bahnasse, Ayoub, Louhab, Fatima Ezzahraa, Oulahyane, Hafsa Ait, Et Al	Nwe Thazin, Khine Moe Nwe, Yutaka Ishibashi
	Approach	real-time measurement system for SDN [33]	MPLS DS-TE Network Management [37]	End-to-end dynamic bandwidth allocation [41]
	Year	2017	2018	2019
Loss	Reference	B. Siniarski, C. Olariu, P. Perry, and J. Murphy	X. Zhang, Y. Wang, J. Zhang, L. Wang, and Y. Zhao	Y. Sinha, S. Vashishth, and K. Haribabu
	Approach	OpenFlow-based VoIP QoE monitoring [34]	A two-way link loss measurement approach [38]	Real time monitoring of packet loss [42]
	Year	2017	2017	2018

procedures are often used. When it comes to vertical handover, these strategies aim to rank the set of potential networks using some assessment characteristics that are structured and meaningful.

To provide a comprehensive overview, we begin by mentioning the AHP, SAW, and TOPSIS mathematical formulations. A vertical handover decision among competing wireless networks is also outlined in terms of applicable formulations for the issue at hand. MADM SAW, TOPSIS, MEW, GRA, ELECTRE, and the Weighted Markov Chain are among the techniques included in this collection (WMC). We will stick to only three here. To provide a comprehensive overview, we begin by mentioning the AHP, SAW, and TOPSIS mathematical formulations.

Analytic Hierarchy Process (AHP)

It is possible to use the Analytical Hierarchy Process (AHP) to assign numerical weights to various decision alternatives based on how well they meet the criteria for reaching a choice. As the relevance of each characteristic is compared to other qualities, the weights are formed. To generate a decision matrix, the method depends on the expertise of experts and pairs-wise comparisons to try to develop a consistent approach. The following is a concise breakdown of the AHP process:

(1) Step 1: Problem Decomposition

First, the problem is decomposed into a hierarchy that contains three levels: (i) the topmost level consists of the overall objective, (ii) the subsequent level represents the decision factors, and (iii) the alternative solutions are located at the bottom level.

(2) Step 2: Construction of a Pair-Wise Comparison Matrix

In the second phase, a pair-wise comparison matrix is constructed to measure the relative relevance of choice criteria. Network quality of service (QoS) metrics are evaluated pairwise for our unique use cases. It is measured using Saaty's 9-point scale, which translates qualitative assessments into numerical priority. Suppose that attribute A is clearly more essential than B, which is why it received a value of 7, while attribute B received a grade of 1/7. The matrix main diagonal is set to 1 to indicate the relative importance of each parameter concerning the others. For the reasons listed above, the elements that are not on the diagonal have inverse symmetry with the primary diagonal.

To normalize the data, each item is first multiplied by the column weight and then added to the comparison matrix. The normalized comparison matrix is then summarized and divided by the number of items in each row. This gives us the result: Using this weight vector, the user can see how much each criterion weighs against the others.

(3) Step 3: Measurement of Consistency of the Weights

The last row of the pairwise comparison matrix contains the total of each column. The sum row of the resultant matrix is then normalized to produce a normalized comparison matrix. In the end, the CR has been gathered. If the value of CR does not exceed 0.1, then a pair-wise comparison matrix is considered to be consistent. One of the most difficult tasks in MADM is figuring out which normalization process is best for which set of criteria.

$$CR = \frac{IC}{IA}$$

With IC $= \frac{\lambda_{\max} - n}{n-1}$.

Simple Additive Weighting (SAW) Method

One of the most commonly used MADM strategies is SAW, which is used to generate candidate scores and alternative rankings. As an example, we will use "m" to signify the number of potential networks and "n" to symbolize the number of decision characteristics. This will show how the approach works. The A SAW context matrix is thus defined as follows:

$$A_{SAW} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \tag{1}$$

where a_{ij} stands for the candidate network's attribute j with network i. A benefit parameter with a higher value results in a better outcome, whereas a cost parameter with a lower value produces a worse outcome. This holds true for all decision parameters. Our focus is on data rate, security, and bandwidth; but, the costs of transmission, delay, st, and jitter are also taken into consideration. Using the following equations, the context matrix A is normalized to yield a matrix A.

For the benefit criteria:

$$\overline{a}_{ij} = \frac{a_{ij}}{\sum_{i=1}^{m} a_{ij}} \tag{2}$$

For the cost criteria:

$$\overline{a}_{ij} = \frac{\sum_{i=1}^{m} a_{ij}}{a_{ij}} \tag{3}$$

The normalized context matrix $\left(\overline{A}\right)$ is given by:

$$\overline{A} = \begin{bmatrix} \overline{a}_{11} & \cdots & \overline{a}_{1n} \\ \vdots & \ddots & \vdots \\ \overline{a}_{m1} & \cdots & \overline{a}_{mn} \end{bmatrix} \tag{4}$$

But for each traffic profile, W is used to designate a specified weight vector whose members w_j 's reflect the weights of criterion with $j = 1,\ldots,n$. Normalization is done by setting the total of w j's to 1.

$$\mathbf{W} = \left[w_1 \cdots \cdots w_n \right] \tag{5}$$

Then, the generated matrix $\overline{\overline{A}}$ is composed by multiplying each column of the $\overline{A}$ matrix that expresses any given attribute by that attribute's weight. This matrix is given by:

$$\overline{\overline{A}} = \begin{bmatrix} \overline{a}_{11} * w_1 & \cdots & \overline{a}_{1n} * w_n \\ \vdots & \ddots & \vdots \\ \overline{a}_{m1} * w_1 & \cdots & \overline{a}_{mn} * w_n \end{bmatrix} \tag{6}$$

Finally, the SAW model assesses all potential networks and assigns a numerical value to each one. Finally, the network with the greatest score is chosen.

$$S_{SAW} = \sum_{j=1}^{n} \overline{a}_{ij} * w_j \tag{7}$$

Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)

Another popular MADM technique for determining the relative importance of potential replacements is the TOPSIS model. Using the greatest and worst networks as benchmarks, it calculates an index that indicates whether a network is closer to the ideal or worse than the worst-case scenario. This is how the context matrix A_{TOPSIS} is built for the provided set of context (m candidate networks, n decision criteria):

$$A_{\text{TOPSIS}} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \tag{8}$$

where the normalization of attributes is given by:

$$\overline{a}_{ij} = \frac{a_{ij}}{\sqrt{\sum_{i=1}^{m} a_{ij}^2}} \tag{9}$$

The weight-normalized matrix $\left(\overline{\overline{A}}_{\text{TOPSIS}}\right)$ is the result of multiplying each context criterion by the weights assigned to each of their respective classes:

$$\overline{\overline{A}}_{\text{TOPSIS}} = \begin{bmatrix} \overline{a}_{11} * w_1 & \cdots & \overline{a}_{1n} * w_n \\ \vdots & \ddots & \vdots \\ \overline{a}_{m1} * w_1 & \cdots & \overline{a}_{mn} * w_n \end{bmatrix} \tag{10}$$

To identify the best and worst alternatives with respect to a given context criterion (with index), we find the highest and lowest values among all $\left(\overline{a}_{ij}\right)$, respectively. The following equations illustrate this.

$$\begin{aligned} \overline{a}_j^{Best} &= Max\left(\overline{a}_{ij}\right), \quad i = 1, \ldots, m \\ \overline{a}_j^{Worst} &= Min\left(\overline{a}_{ij}\right), \quad i = 1, \ldots, m \end{aligned} \tag{11}$$

Finally, the distances between the best and worst solutions are calculated for each candidate network as follows:

$$\begin{aligned} S_i^+ &= \sqrt{\sum_{j=1}^{n} \left(\overline{a}_{ij} - \overline{a}_j^{Best}\right)^2} \\ S_i^- &= \sqrt{\sum_{j=1}^{n} \left(\overline{a}_{ij} - \overline{a}_j^{Worst}\right)^2} \end{aligned} \tag{12}$$

S_{TOPSIS} is used to find the optimal option by comparing the distances between the best and worst answers.

$$S_{\text{TOPSIS}} = \frac{S_i^-}{S_i^+ + S_i^-} \tag{13}$$

Secure Model for Dynamic Access Control and Unreliable Access Point Detection

In this section, we describe the architecture of our proposed model. Figure 2 describes the different components of our architecture. The proposed architecture consists of a set of three layers: the application layer, the control layer, and the data layer. The details for these layers will be discussed in the following subsections.

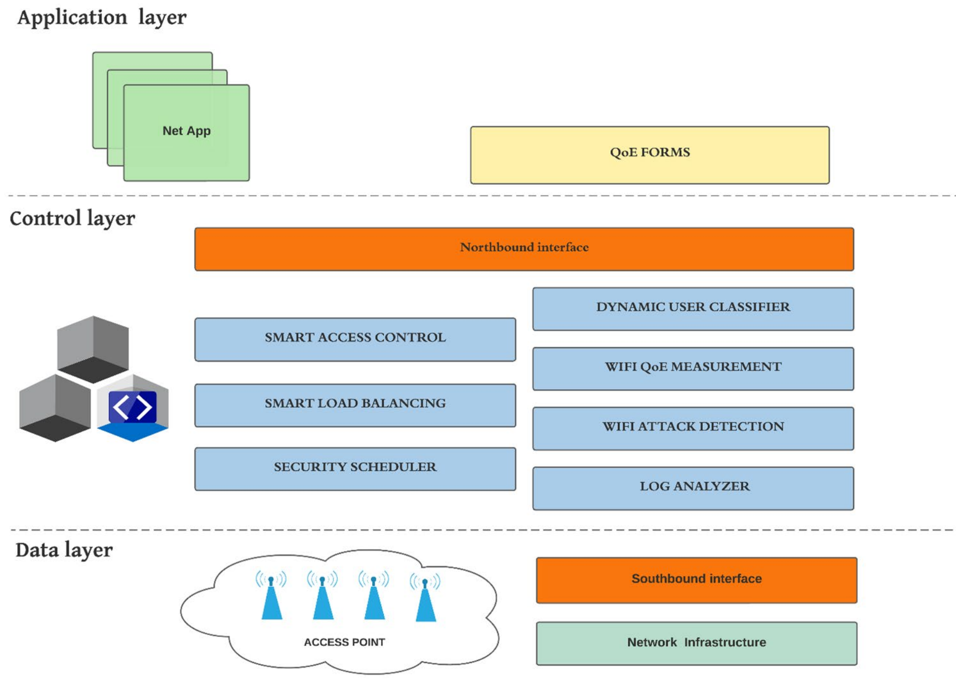
The Application Layer

This layer is in direct relation with the user, it allows him to configure, manage and secure the resources. It groups all the applications that can be used directly by the users.

QoE Forms

The most effective way to monitor network performance as close as possible to the user's experience is to measure the quality of experience (QoE). To achieve this, we thought of creating a form to track user feedback.

Fig. 2 SADMQOS architecture plans



This form is displayed after a dT (the average time of connection to an access point) so that the user can give his opinion on the quality of the network.

To increase the efficiency of the questionnaire, personalized questionnaires have been created, and adapted to the nature of the user.

Below are two examples of surveys (Fig. 3a and b). For experienced users' survey, the questions are more specific than for a simple user's survey. The goal is to use the expertise of these employees to the benefit of our system. The second question is about the stability of the connection: whether the access point has experienced recurring small signal drops or the throughput has had a significant drop at some point. Regarding the question about threats perceived by the user, here are some examples: when the user processes confidential data and suddenly the HTTPS logo is no longer displayed it means that the browser no longer establishes a secure connection and that perhaps a hacker is trying to spy on the user. Again, if the firewall blocks or prevents data transmission then the source is not reliable. There are also notifications from the anti-virus software that will periodically filter and inform the user if there are any problems.

The Control Layer

This layer is the layer responsible for managing the policies and the flow of traffic within the network and it also deals with the organization of the applications' demand and the implementation of the rules.

Security Scheduler

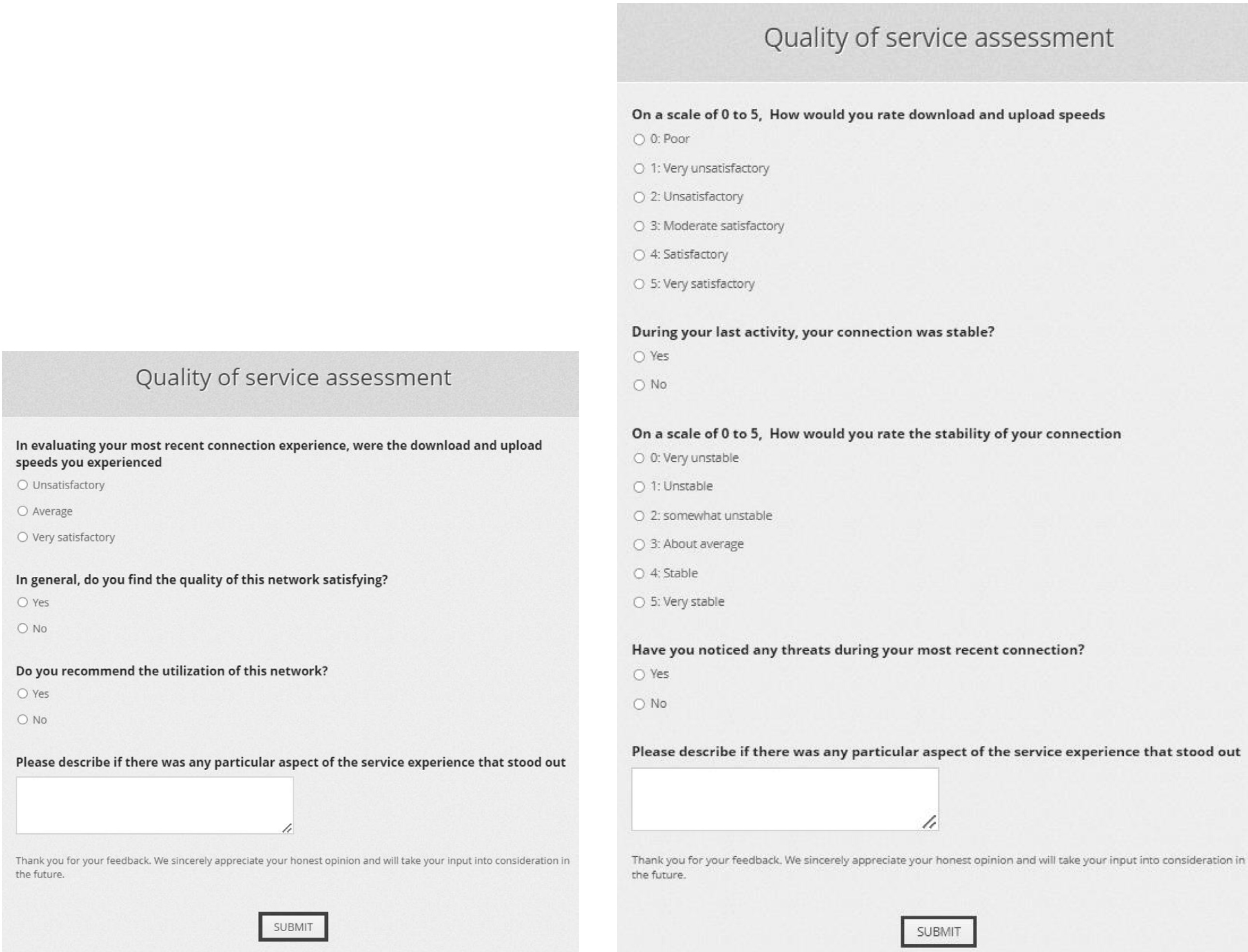
This module provides a centralized view of the modules of the same level as well as the synchronization of their outputs.

Our architecture performs the following tasks:

- The controller creates an access point that has the highest RSSI
- After authentication, perform access control based on mac address and geolocation
- If the user is not a priority (does not require security): connect to the appropriate access point
- If not, we retrieve the log file, analyze the messages, and detect attacks.
- Perform network performance monitoring using the form.
- Choose the most suitable access point for the requirements

The main authentication scenario is detailed in Fig. 4.

Figure 5 summarizes the tasks described above, the process of requesting the connection involves the collaboration of six modules between them. This request goes through several tests to verify if the choice of access point meets the requirements of the user.



(a) Survey for a simple user (b) Survey for experienced users

Fig. 3 Examples of survey

Smart Load Balancing

Smart Load Balancing is responsible for load balancing for a fair and optimal distribution of users between the access points and the efficiency of resource use.

This module is the continuation of the two modules Smart access control and Wifi attack detection. As a first step, this module retrieves a list of users with their constraints and the available access points.

The table maps the user classifications to the weight of the available access points. To create this table, it is essential to classify and give the weight of each access point according to its throughput, the number of hosts to connect, and the remaining throughput. This table will manage the network flow using the SD controller of the OpenFlow network switches. Figure 6 resumes the processes of smart load balancing.

Algorithm 1: Load_Balancing_Type This algorithm determines the type of load balancing to be used based on the load list of the controller. It compares the loads of different resources and returns 1 if the costs are unequal and 0 if they are equal. The output of this algorithm is used as input for Algorithm 2.

Algorithm 2: Load Balancing This algorithm performs load balancing based on the output of Algorithm 1 (load_balancing_type). If the load balancing type is determined to be unequal cost (var = true), it applies the RoundRobin algorithm (Algorithm 3) to distribute the workload among the resources. If the load balancing type is equal cost (var = false), it follows a different logic, iterating through the access points (Apis) and assigning users to them based on their preferences and the calculated number of users. This algorithm ensures that the workload is distributed effectively based on the load balancing type.

Fig. 4 Authentication scenario

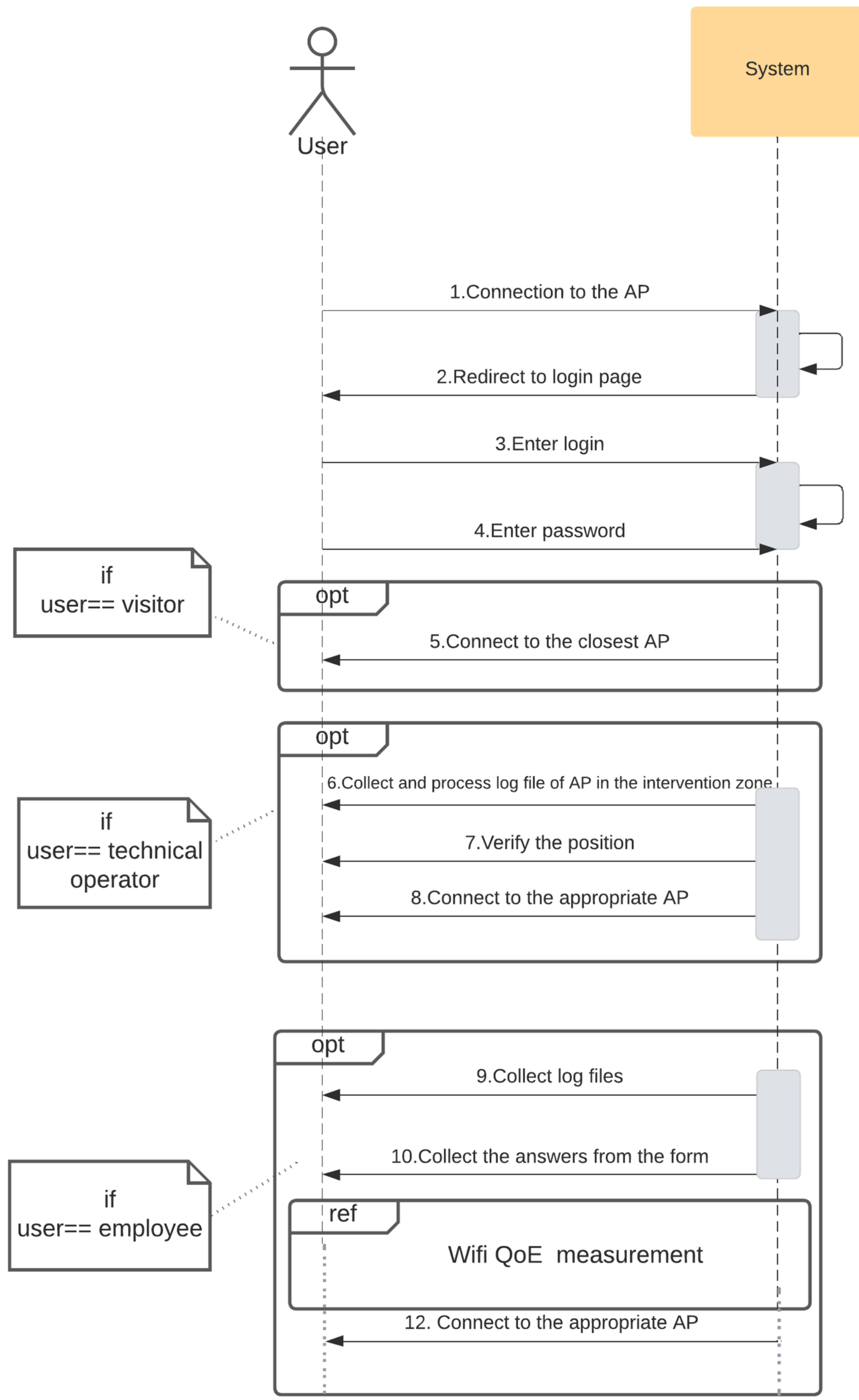
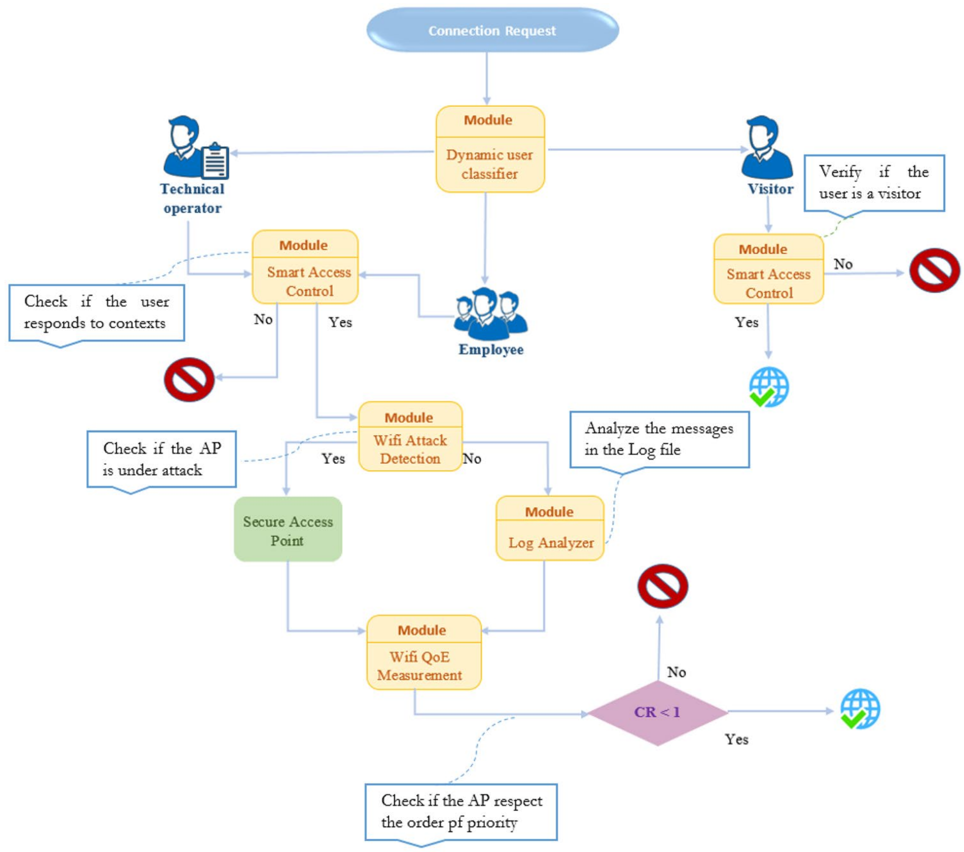


Fig. 5 Request connection process



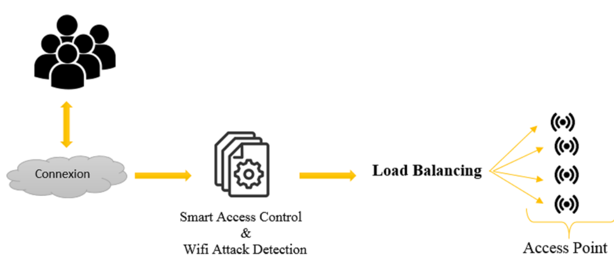


Fig. 6 Processes of smart load balancing

Algorithm 3: Round Robin This algorithm is a classic scheduling algorithm that aims to evenly distribute tasks among resources over time. It takes a task queue and a quantum (a fixed time interval) as input. It then schedules tasks in a round-robin fashion, allocating a fixed time slice (quantum) to each task before moving to the next task. This ensures that each task gets an equal share of processing time. The output of this algorithm is a schedule list that represents the order in which tasks are scheduled.

These three algorithms are related to load balancing in a network environment. Algorithm 1 determines the load balancing type, Algorithm 2 performs load balancing based on the type determined by Algorithm 1, and Algorithm 3 (Round Robin) is used as a load balancing method within Algorithm 2 when the load balancing type is unequal cost. Together, these algorithms work collaboratively to achieve load balancing and optimize resource utilization in the network environment.

Api: Access Point i.

Wi: Weight of Api.

input: Controller's load list

output 0: equal cost

1: unequal cost

for w=0 to n w++ **do**

if w_i != w_{i+1} then

return 1

}

return 0;

endFor

Algorithm 1 load_balancing_type

```
INPUT:   user's list; load_balancing_type ()
//  we will use the output of the previous function
OUTPUT: succeful load balancing
var= load_balancing_type ();
IF (var = true)
    Apply RoundRobin ()
ELSE
    for Ap = 0 to n
        get Preference of Api
        x= calculate number of users
        For U = temp to x
            Api = Uj;
            temp = j;
ENDIF
END
```

Algorithm 2 load balancing

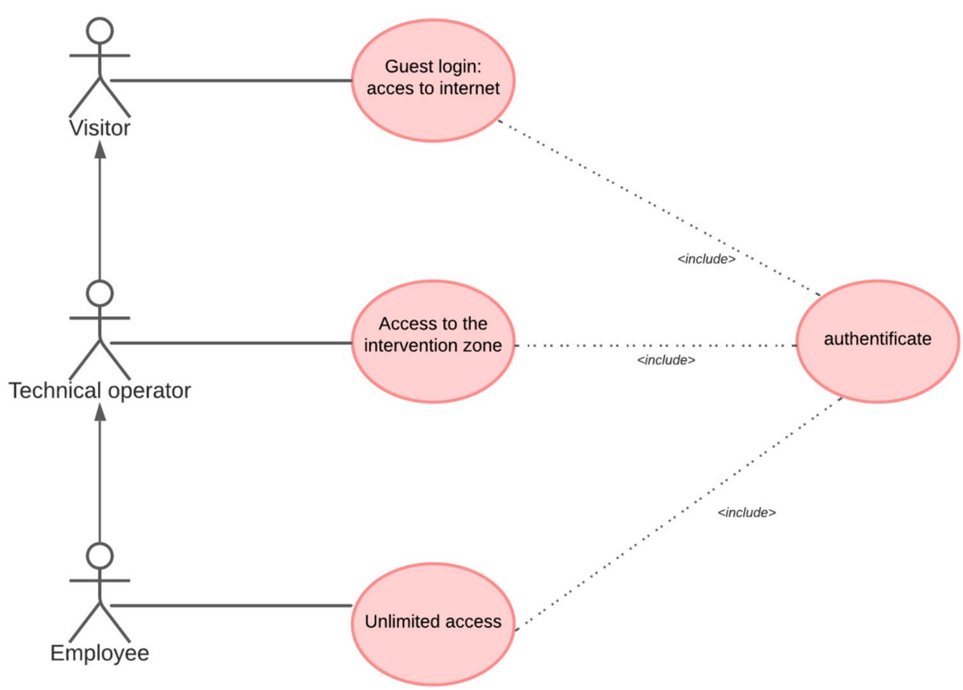
```
INPUT: task_queue, quantum
OUTPUT: scheduled tasks
Create an empty list, "schedule"
Set a variable "time" to 0
While task_queue is not empty:
Get the next task from task_queue
 If the remaining time of the task is greater than quantum: THEN
    Deduct quantum from the remaining time of the task
    Add quantum to the value of "time"
    Append the task to the "schedule" list
  Else:
    Deduct the remaining time of the task from the value of "time"
    Remove the task from the task_queue
    Append the task to the "schedule" list
  ENDIF
Return the "schedule" list
END
```

Algorithm 3 Round Robin

Pi: Preference of Api.
n: Number of access points.
Advantages of Round Robin Scheduling:

- *Fairness*: Round Robin provides fairness by allocating CPU time equally among all processes. No single process monopolizes the CPU, ensuring that each process gets a chance to execute regularly.
- *Response Time*: Round Robin scheduling guarantees a quick response time for interactive applications. Since each process gets an equal share of CPU time, even long-running processes do not delay the response of other processes.
- *Real-Time Systems*: Round Robin is often used in real-time systems where predictability and time constraints are critical. The fixed time quantum allows for better

Fig. 7 User use cases



```
INPUT P
OUTPUT: LOG message
SWITCH (P)
    case 0-2:
        a = a + 1
        break
    case 3-4:
        b = b + 1
        break
    case 5-7:
        c = c + 1
        break
    default:
        print("error")
IF (a > b) and (a > c):
    return 0
ELSE
    return 1
ENDIF
```

Algorithm 4 Fichier Log

predictability and guarantees that every process gets a chance to execute within a specific timeframe.

- *Multitasking*: Round Robin is an efficient algorithm for multitasking environments. It enables simultaneous execution of multiple processes, making it suitable for environments with concurrent user interactions or multitasking operating systems.
- *Easy Implementation*: Round Robin is relatively simple to implement compared to other scheduling algorithms. It does not require complex priority calculations or resource dependencies, making it easier to understand and manage.

Dynamic User Classifier

This module allows the user to authenticate himself, according to our architecture the user whatever his function starts by connecting to the access network then a session opens where he must be identified using his username and password.

Figure 7 describes all the user use cases that require authentication.

Smart Access Control

Roles This module allows us to optimize the use of our model. By distinguishing the different types of users and assigning access to those who need it. There are three types of users:

- Visitors are users who only use the network for personal reasons, for a short time. They have basic permissions.
- Technical operators are users who have access only in their area of intervention, and during working hours. They are assigned permissions according to their needs.
- Employees are permanent users who have access within their department, and during working hours. They are assigned permissions according to their responsibilities within the organization.

Activities and Actions The actions that users can perform within the organization are listed below:

- Administer: create, delete, allow, and disallow.
- Execute: download.
- Modify: write.
- Consult: read, search.

Contexts *Spatial contexts*: administration, dep_physics, dep_chemistry, dep_maths, dep_biology, and dep_geology.

- VsVoVa (Define (Fac, s, a, o, dep_physics) $ (@ IP = 192.168.90.0/24)): this means that the context dep_physics is valid if only ip address of the user belongs to the network 192.168.90.0/24
- VsVoVa (Define (Fac, s, a, o, dep_chemistry) $ (@ IP = 192.168.128.0/24)): this means that the context dep_chemistry is valid if only ip address of the user belongs to the network 192.168.128.0/24

```
AttackAlert (Gn, Xn)
INPUT: Gn, Xn, h, t
OUTPUT: Attack message
Set Gn = 0
WHILE (Gn < h) do
    FOR n = 0 to t
        µn = 0.5 * µn-1 + 0.5 * Xn-1
    ENDFOR
Gn = Gn-1 + (Xn - µ)
ENDWHILE
Display attack message
END
```

Algorithm 5 Traffic Attack

- VsVoVa (Define (Fac, s, a, o, dep_ maths) $ (@ IP = 192.168.128.0/24)): this means that the context dep_ maths is valid if only ip address of user belongs to the network 192.168.128.0/24
- VsVoVa (Define (Fac, s, a, o, dep_ biology) $ (@ IP = 192.168.128.0/24)): this means that the context dep_ biology is valid if only ip address of the user belongs to the network 192.168.128.0/24
- VsVoVa (Define (Fac, s, a, o, dep_ geology) $ (@ IP = 192.168.128.0/24)): this means that the context dep_ geology is valid if only ip address of the user belongs to the network 192.168.128.0/24
- VsVoVa (Define (Fac, s, a, o, administration) $ (@ IP = 192.168.128.0/24)): this means that the administration context is valid if only ip address of the user belongs to the network 192.168.128.0/24

Temporal contexts: depending on the nature of the role, we can define three types of timing.

- VsVoVa (Define (Fac, s, a, o, full_time) $ (h > 8 ^ h B 12) v (h > 14 ^ h < 18)): this means that the full_time context is valid if only the time of the request belongs to the normal time interval.
- VsVoVa (Define (Fac, s, a, o, continious_time)$ (h > 8 ^ h < 13))): this means that the context continious_time is valid if only the query time belongs to the continuous time interval.
- VsVoVa (Define (Fac, s, a, o,cnx_p)$ (p < 120 m)): this means that the context cnx_p is valid if the user does not exceed two hours of connection.

Compound contexts: this context combines two types of spatial and temporal contexts to handle some exceptional cases.

- VsVoVa (Define (Fac, s, a, o, intervention_ctx) $ (dep_ chemistry ^ cnx_p)): this means that the context is valid if both contexts are valid: so this case the context is true if the user belongs to the chemistry department and does not exceed 2 h of connection.

Log Analyzer

f to analyze the messages in the Log file and classify them according to their type and priority. The following table describes the different messages from zero to seven where 0 = emergency message and 7 = debugging message.

Wi-Fi Attack Detection

This module consists of detecting access points that have been attacked, for this we worked on active attacks and given their large number we worked on the three most common attacks on the access point side: Dos, Rogue AP, and Disassociation.

- DoS (Denial of Service) aims to prevent legitimate users from accessing services by saturating those services with false requests. It is generally based on software "bugs". In a wifi network, this consists in particular in blocking access points either by flooding them with de-association or de-authentication requests
- Rogue AP (rogue access point) is a technique that consists in installing a wireless access point in a secured network without having authorization from the administrator or the network manager. This attack can cause several damages such as data theft or destruction and loss of information, sharing of erroneous or malicious data, forced shutdown of some services, third party attacks …
- De-association of clients: an access point can signal to a client that it has lost its association and that it must re-authenticate. Similarly, a client must signal the access point that it is leaving the network. It is then quite easy for an attacker to create de-association frames and to constantly force a machine to negotiate its connection. The latter would then spend more time associating than sending useful data.

1.INPUT: APs, authorized APs

2. OUTPUT: Rogue APs

3. Create an empty list, "RogueAPs"

4. For each AP **in** APs:

5. Check if the AP is in the authorized APs list

6. **ELSE**

7. add the AP to the "RogueAPs" list

8. Return the "RogueAPs" list

9.END

Algorithm 6 Rogue AP Detection

We have implemented an algorithm that detects anomalies caused by the mentioned attacks: a warning message is triggered if the number of packets sent to the network exceeds the global traffic threshold and in case the connection and disconnection of the same machine are repeated in a very short time.

Gn is the sum, which is initialized to zero when the algorithm is launched.

Xn is the variable representing the number of packets.

μ is the average of Xn calculated at time t (n going from 0 to t).

Algorithm 5: Traffic Attack This algorithm is used to detect traffic attacks in a network environment. It takes inputs Gn (current traffic value), Xn (incoming traffic value), h (threshold value), and t (time interval). The algorithm iterates until the traffic value (Gn) reaches the specified threshold (h). Within each iteration, it calculates the weighted average of the previous traffic value ($\mu n-1$) and the previous incoming traffic value ($Xn-1$). It then updates the current traffic value (Gn) based on the difference between the incoming traffic value (Xn) and the calculated average (μ). Once the threshold is reached, an attack message is displayed.

Algorithm 6: Rogue AP Detection This algorithm is used to identify rogue access points (APs) in a network. It takes inputs APs (list of detected APs) and authorized APs (list of trusted and authorized APs). The algorithm creates an empty list called "RogueAPs" to store any detected rogue APs. It iterates through each AP in the APs list and checks if it exists in the authorized APs list. If an AP is not found in the authorized APs list, it is considered a rogue AP and added to the "RogueAPs" list. Finally, the algorithm returns the list of identified rogue APs.

Data Integrity Verification

This module will be responsible for verifying the integrity of the received information in the wireless network. It would ensure that the data packets are not tampered with or modified during transmission.

By incorporating the Data Integrity Verification module, you can add an additional layer of security and assurance to your system, ensuring that the correct and unmodified information is received. This module helps to detect any potential tampering or data corruption that may occur during the transmission process.

Wifi QoE Measurement

This module allows the user to monitor network performance as closely as possible to the user experience. It allows classifying the networks according to their quality of service, availability, reliability, and security.

Based on the information stored in the QoE forms module, we can study the state of the connected access points over a time interval.

We will use the AHP method, which is a model based on the comparison of pairs of factors at the same hierarchical level. For this methodology, we have used four criteria: reliability, throughput stability, security, and throughput. The first step is to develop a binary comparison matrix for each criterion. The weights assigned to each criterion come from

	1	2	3	4
1	1	0.33	3.00	0.25
2	3.00	1	8.00	1.00
3	0.33	0.12	1	0.14
4	4.00	1.00	7.00	1

Fig. 8 The decision matrix

Cat		Priority	Rank	(+)	(-)
1	Fiabilité	12.9%	3	1.9%	1.9%
2	Stabilité débit	40.0%	2	1.4%	1.4%
3	Sécurité	5.1%	4	0.7%	0.7%
4	Débit	42.0%	1	6.8%	6.8%

Fig. 9 Order of criteria and their priorities

an understanding of the needs and nature of the users. In addition, to ensure the reliability of this method, we calculate the consistency ratio (CR) using the following formula:

$$CR = \frac{IC}{IA}$$

With: $IC = \frac{\lambda max - n}{n-1}$

λ max: maximum eigenvalue of each factor in the matrix array, n: the size of the matrix, IA: random index of consistency (SAATY).

The value of CR should not exceed 10% ~ CR ≤ 0.1 to show that the matrix is consistent (Fig. 8).

Case 1: Simple user.

Figure 9 represents the order of the criteria and the priority of each criterion.

CR = 0.9%

Case 2: Experienced user (Fig. 10).

Figure 11 represents the order of the criteria and the priority of each criterion.

CR = 0.8%.

Results and Discussion

In our study, we conducted various scenarios to assess the scalability of our model. To achieve this, we increased the number of users while limiting the available bandwidth. We utilized the UNIFI tool, which supports the new SDN technology and provides a fully integrated platform for our experiments. To enforce bandwidth limitations, we utilized the Bandwidth Profiles option, allowing us to configure download and upload bandwidth limits and assign them to specific users.

We compared the performance of our model against the basic FIFO model and a scenario without any QoS provisions. Our objective was to evaluate the performance of our model across different types of traffic, including VoIP, HTTP, and FTP, based on several criteria:

	1	2	3	4
1	1	0.50	0.17	2.00
2	2.00	1	0.25	3.00
3	6.00	4.00	1	9.00
4	0.50	0.33	0.11	1

Fig. 10 The decision matrix

Cat		Priority	Rank	(+)	(-)
1	Fiabilité	10.7%	3	1.3%	1.3%
2	Stabilité débit	18.6%	2	2.2%	2.2%
3	Sécurité	64.5%	1	7.6%	7.6%
4	Débit	6.2%	4	0.7%	0.7%

Fig. 11 Order of criteria and their priorities

- *Latency (End-to-End Delay)*: This measures the time required for a data packet to be transmitted from the sender to the recipient. It is a critical metric in assessing network performance.
- *Jitter*: Jitter refers to the variation in latency or the difference in transmission time between packets sent from one system to another within a network. It can impact the overall quality of service.
- *Packet Loss*: Packet loss is the percentage of packets that are lost during transmission. While some loss can be tolerated, it should be kept within acceptable limits, typically below 3%, to avoid significant degradation in audio quality.
- *MOS (Mean Opinion Score)*: MOS provides a numerical measure, ranging from 1 to 5, that quantifies the audio quality at the destination end. Higher scores indicate better audio quality.
- *HTTP Transaction Time*: This metric measures the time required to complete an HTTP transaction, which includes the stages of connection establishment, request, response, and closure.
- *HTTP Delay*: HTTP delay specifically refers to the time required to configure the connection and transfer the response request. Since HTTP uses TCP as the transport protocol, this delay is influenced by the performance of TCP.

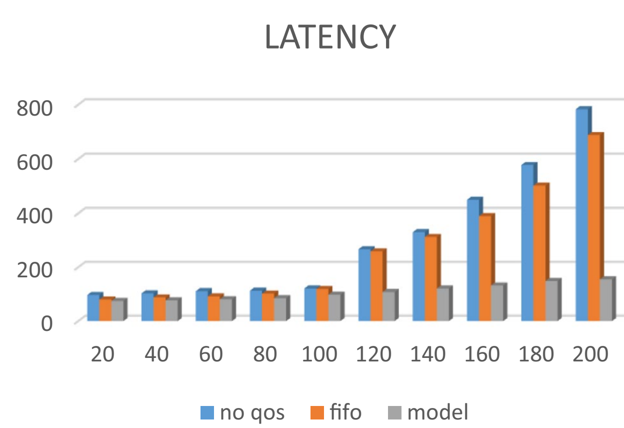


Fig. 12 Voip latency

By analyzing these performance indicators in different scenarios, we aim to assess the effectiveness of our model in providing improved latency, reduced jitter, minimal packet loss, satisfactory MOS scores, and efficient HTTP transaction times.

Figure 12 illustrates the latency, also known as end-to-end delay, in the three scenarios. In wireless networks, latency can be higher compared to wired networks due to various factors such as distance, propagation delay, router quality, packet size, and available bandwidth. Higher latency values negatively impact network performance.

VoIP industry leaders like CISCO have defined a maximum tolerable latency value of 150 ms. The results obtained from the scenarios demonstrate that both the No QoS and FIFO models exhibit significantly high latency values, surpassing the threshold. In contrast, our proposed model achieves lower latency values.

Even with an increasing number of users, our model maintains latency below 130 ms. In comparison, the FIFO model reaches 350 ms, and the No QoS model experiences latency exceeding 450 ms. This improvement in latency is attributed to the smart utilization of traffic prioritization in our model, with VoIP traffic classified as high priority.

By prioritizing VoIP traffic and optimizing resource allocation, our model effectively reduces latency, ensuring a more responsive and efficient network performance compared to the No QoS and FIFO scenarios.

Figure 13 depicts the jitter observed in the three scenarios: No QoS, FIFO, and our proposed model. Jitter, as mentioned earlier, refers to the variation in latency and can be caused by factors such as network congestion and hardware issues.

It's important to note that an increase in latency does not necessarily indicate an increase in jitter. Jitter primarily arises from the continuous variation in latency. By comparing the results obtained with the standards defined by CISCO, it is recommended to keep jitter below 50 ms. Unfortunately, the No QoS and FIFO scenarios failed to meet this requirement.

In our proposed model, thanks to the dynamic allocation of resources, high-priority classes are able to utilize any unused bandwidth from lower-priority classes. Once released, our model reallocates this bandwidth to the classes that require it. This dynamic resource allocation mechanism helps to mitigate the impact of jitter, ensuring that it remains within acceptable limits.

The results obtained from our model demonstrate its effectiveness in managing jitter and meeting the recommended standards defined by CISCO. By efficiently allocating and reallocating resources, our model provides a more stable and reliable network performance compared to the No QoS and FIFO scenarios.

In wireless networks, data loss issues are more prevalent compared to wired networks. Figure 14 illustrates the packet loss rate in the three scenarios. It is evident that as the number of users increases, the number of packets sent also increases, leading to a greater likelihood of some packets not reaching their intended destination. This phenomenon directly contributes to the degradation of quality of service.

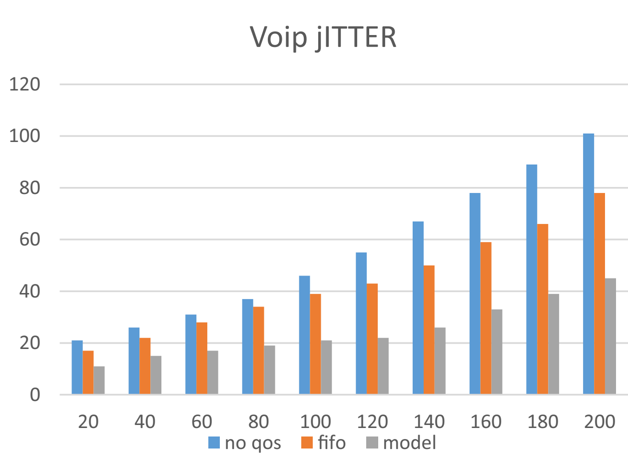


Fig. 13 Voip jitter

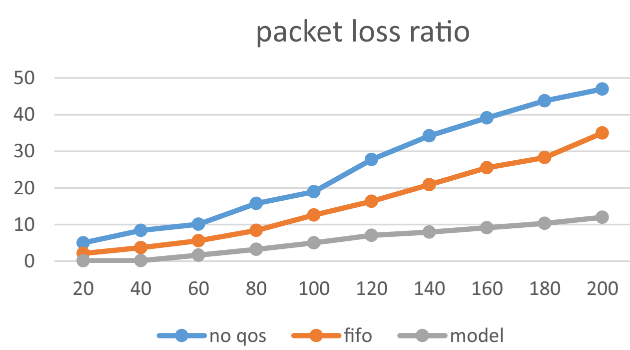


Fig. 14 Packet loss ratio

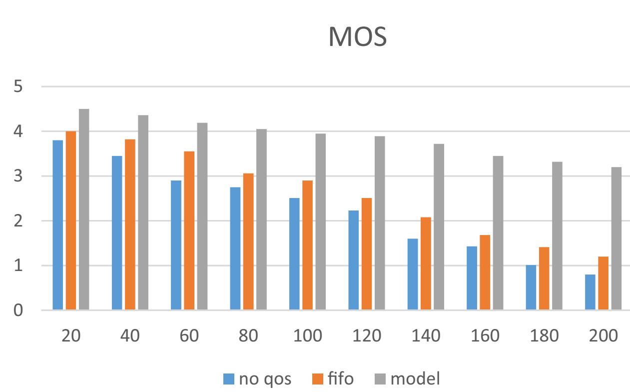


Fig. 15 Mean opinion score

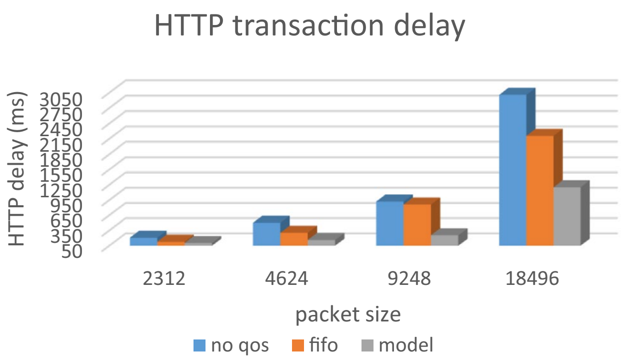


Fig. 16 HTTP transaction delay

Packet loss can occur due to various factors, including network congestion, weak signals, radio frequency interference, and hardware problems. It is important to note that maintaining an optimal packet loss rate is crucial for ensuring quality voice traffic. According to CISCO, a packet loss rate of up to 1% is considered optimal for voice traffic, while a rate of up to 3% is tolerable.

The observed packet loss in the scenarios indicates the impact of these factors on the quality of service in wireless networks. By implementing appropriate QoS mechanisms, such as our proposed model, it is possible to mitigate the effects of packet loss and maintain a satisfactory level of service quality, even in the presence of increasing user loads.

In Fig. 15, we present the average opinion score (MOS) obtained during VoIP calls in the three scenarios. MOS is a subjective assessment that evaluates the audio and visual quality of VoIP traffic. It takes into consideration factors such as bandwidth, jitter, data loss, and latency. The MOS is rated on a scale from one to five, with higher values indicating better quality.

In our study, we observed that our proposed model consistently achieved an acceptable MOS, even when the number of users increased up to 200. The MOS score for our model remained at 3.2, indicating a satisfactory level of quality. In contrast, the FIFO scenario obtained a MOS score of 1.2, and in the absence of QoS provisions, the MOS score dropped even further to 0.8.

These results demonstrate the effectiveness of our model in ensuring acceptable audio and visual quality during VoIP calls, even under high user load. By effectively managing bandwidth, reducing jitter and data loss, and minimizing latency, our model significantly outperforms the traditional FIFO approach and the scenario without QoS provisions.

In Fig. 16, we observe the delay experienced by HTTP transactions. Since HTTP utilizes TCP as its underlying transport protocol, the delay variation is influenced by the performance of TCP. The transaction begins with the establishment of the connection, during which no packets are sent until the connection is established. This initial step explains the increase in HTTP delays.

In our evaluation, we found that the FIFO model achieved optimal delays for up to 9248 packets. Beyond this point, the model deviated from the recommended rate specified by the ITU-T, exceeding the optimal value of 2 s. On the other hand, our proposed model demonstrated its efficiency by reaching delays of up to 18,496 packets. Remarkably, our model did not surpass 1200 ms, whereas the FIFO model reached 2207 ms and the no QoS case resulted in delays of 3010 ms.

These results highlight the superiority of our model in managing delays for HTTP transactions. By optimizing the performance of TCP and implementing QoS mechanisms, our model successfully ensures lower delays, providing a more efficient user experience compared to the traditional FIFO model and the absence of QoS provisions.

Conclusion

The advent of Software-Defined Networking (SDN) technology has brought about a significant revolution in the field of networking. It is based on the principle of separating the control plane and the data plane, introducing a centralized controller for managing and controlling network flows. This novel approach has not only proven successful in wired networks but also in wireless networks, effectively addressing the needs of modern users. However, despite its successes, several challenges persist, including resource allocation, latency, high costs, and data rate limitations. In light of these challenges, our focus in this study is on Quality of Service (QoS) management in wireless networks.

In this paper, we propose a secure model that incorporates dynamic access control and the detection of unreliable access points. Additionally, we present a QoS management model for wireless networks by leveraging the SDN approach, aimed at addressing connectivity issues and optimizing resource utilization. To validate our model, we conducted several scenarios under different conditions and implemented various types of traffic, including VoIP, HTTP, FTP, and video streaming. The results of our model's implementation demonstrated its efficiency when compared to the traditional First-In-First-Out (FIFO) model, particularly in scenarios with no QoS provisions. Our model offers advantages such as reduced latency and jitter, satisfactory average opinion scores, minimal packet loss, and acceptable transaction delay.

Looking ahead, future work will concentrate on exploring link aggregation techniques to further enhance performance. Additionally, we plan to strengthen the security aspects of our model to ensure a robust and resilient system.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest in this work.

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